

- I) Hydrocarbons That Had To Form In Place
 - A) Hydrocarbon Source Rocks That Are Now Reservoir Rocks (Low Permeability)
 - B) Isolated Sand Lenses - Completely Encased In Shales
- II) Migrated Hydrocarbons That Come From Tight Source Rocks - That Still Contain Large Volumes of Hydrocarbons
- III) Major Sand Injectites and Re-Mobilization
 - A) Associated With Hydrocarbons
 - B) Large Sand Injectites Not Directly Associated With Oil and Gas.
 - C) Mud Volcanoes
- IV) Long Distance Lateral Migration
- V) Large Vertical Migration From Deep Sources
- VI) Hydrocarbons Sourced By Too Ancient of Source Rocks
- VII) Evidences of Early Oil Migration
 - A) Diagenetic Traps (Q & K decrease below O/W)
 - B) Compaction Inhibition From Early Hydrocarbon Migration
 - C) Bacterial Tar Mats
 - D) Closed Systems - Isolated From Migration Paths.
 - E) Traps That Are Barren Only Because Of Late Trap Formation (Near To Older Productive Traps)

VIII - Proofs of Fast HC Migration And Diffusion Rates

A) Oil And Gas Seeps

B) Hydrocarbon Microseepage & Geochemical Exploration

C) Diffusion Models And Rates

IX - Proofs of Rapid Formation of Oil

A) Laboratory Pyrolysis

B) Renewable (Synthetic)

C) Recent Sediments

D) Hydrothermal

my Thesis: After more than 150 years of Hydrocarbon Exploration and Production, the petroleum industry still does not have a reasonable explanation of Primary Migration (Hydrocarbon migration out of the source rocks into carrier beds).

my Thesis: The totality of data available both, geological and engineering, fits better into a GFS (Global Flood Scenario) or RCS (Rapid Catastrophic Sedimentation Scenario) than ~~a~~ ^{the current} ~~convention~~ Old Earth (uniformitarianism) ^{15m} where most processes are slow and exactly what we see today.

Oil Window: 60°C to 120°C (140°F to 250°F)
2 to 4 KM (6500 ft to 13000 ft)

Early Oil: 60°C to 90°C (140°F to 200°F)

Peak oil: 90°C to 120°C (200°F to 250°F)

Late Oil: 120°C to 150°C (250°F to 300°F)

Early Gas

Gas Window: $>150^{\circ}\text{C}$ ($>300^{\circ}\text{F}$)

2008 Paper (Energy Exploration & Exploitation) Shengjun Qin
Showed immature generation of oil in 5 years at 80°C (170°F)
Equivalent to the Early Oil Window